

A TWO-LEVEL DIGIT LAW FOR APÉRY'S PAIR

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ABSTRACT. Let a_n be the Apéry numbers for $\zeta(3)$ and b_n their companion. We show that the pair $(a_n, p^3 b_n)$ transforms under base- p digit splitting by a single upper-triangular matrix whose diagonal is one explicit scalar and whose off-diagonal entry is the low digit's own companion value, scaled by exactly p^3 . The scalar is the second-order truncation of a Γ -deformation of the summand, summed over the full digit range. The first-order part of the law is proved; it is rank one, and we prove the identity that makes it so.

Throughout p is a prime ≥ 5 , and for $x, y \in \mathbb{Q}$ we write $x \equiv y \pmod{p^m}$ to mean $v_p(x - y) \geq m$. Evidence is labelled: statements called Theorem, Lemma or Proposition are proved; those called Observation are exact computations over the stated range; Conjectures are neither.

1. THE PAIR, AND THE TWO FUNCTIONALS

Definition 1.1. $A(n, k) = \binom{n}{k}^2 \binom{n+k}{k}^2$, $a_n = \sum_{k=0}^n A(n, k)$, $H_m^{(r)} = \sum_{j=1}^m j^{-r}$, $H_m = H_m^{(1)}$, and $b_n = \sum_{k=0}^n A(n, k)(2H_n^{(3)} - H_k^{(3)})$, the solution of

$$(n+2)^3 b_{n+2} = (34n^3 + 153n^2 + 231n + 117) b_{n+1} - (n+1)^3 b_n, \quad b_0 = 0, \quad b_1 = 6.$$

Definition 1.2. $u(r, s) = H_{r+s} - H_{r-s}$ and $v(r, s) = H_{r+s} + H_{r-s} - 2H_s$; these are $\frac{1}{2}\partial_n \log A$ and $\frac{1}{2}\partial_k \log A$. Put

$$U_r = \sum_s A(r, s) u(r, s), \quad V_r = \sum_s A(r, s) v(r, s).$$

$U_0, \dots, U_8 = 0, 6, 105, 2219, \frac{104825}{2}, \frac{13276637}{10}, \frac{70543291}{2}, \frac{67890874657}{70}, \frac{766399019471}{28}$. Note $U_r = \frac{1}{2} da_n/dn$ in the Γ -form.

2. THE EXPANSION, AND WHY THE DEFECT IS RANK ONE

Lemma 2.1 (digit expansion). *For $0 \leq s \leq r < p$, $0 \leq c \leq a < p$ and $r + s < p$,*

$$A(ap + r, cp + s) \equiv A(a, c) A(r, s) (1 + 2p[au(r, s) + cv(r, s)]) \pmod{p^2}.$$

Outside that region the relevant binomial is $\equiv 0 \pmod{p}$; since A is a square, both $A(ap + r, cp + s)$ and $A(r, s)$ are then $\equiv 0 \pmod{p^2}$, so those regions do not contribute at first order at all.

Proof. Only the weak Wolstenholme congruence $H_{p-1} \equiv 0 \pmod{p}$ is needed (the $(ip)^2$ terms die of their own accord), and it follows from pairing $l \leftrightarrow p - l$. It makes the $(ap)!/(p^a a!) \equiv ((p-1)!)^a$ factors cancel, giving $\binom{ap+r}{cp+s} \equiv \binom{a}{c} \binom{r}{s} (1 + p[au(r, s) + cv(r, s)])$ and $\binom{(a+c)p+(r+s)}{cp+s} \equiv \binom{a+c}{c} \binom{r+s}{s} (1 + p[(a+c)H_{r+s} - cH_s - aH_r])$ modulo p^2 . In the product A the two aH_r terms cancel. $\square$

Lemma 2.1 is available only on $r + s < p$, so summing it produces the *restricted* functionals $\tilde{U}_r = \sum_{s \leq r, r+s < p} A(r, s) u(r, s)$ and $\tilde{V}_r$, not U_r and V_r . The two agree:

Lemma 2.2. *For $r < p$, $\tilde{U}_r \equiv U_r$ and $\tilde{V}_r \equiv V_r \pmod{p}$.*

Proof. For $r+s \geq p$ the carried binomial satisfies $\binom{r+s}{s} \equiv 0 \pmod{p}$, and A is a square, so $p^2 \mid A(r, s)$; while $u(r, s), v(r, s)$ have arguments $< 2p$ and hence $v_p \geq -1$. Every dropped term therefore has $v_p \geq 1$. $\square$

This is not a formality: U_r is a p -independent rational which *does* carry p in its denominator for small r — $U_5 = 13276637/10$ and $U_7 = 67890874657/70$, so $v_5(U_5) = v_7(U_7) = -1$. It is p -integral precisely because $r < p$, which is a consequence of Lemma 2.2 rather than of anything stated before it.

Since $H_{ap+r}^{(3)} = p^{-3}H_a^{(3)} + G_{ap+r}$ with $G_m = \sum_{j \leq m, p \nmid j} j^{-3}$ p -integral, the weight contributes nothing below order p^3 . Lemma 2.1 therefore gives, with $a'_a = \sum_c c A(a, c)$ and $b'_a = \sum_c c A(a, c)(2H_a^{(3)} - H_c^{(3)})$,

$$\frac{1}{p}(a_{ap+r} - a_a a_r) \equiv 2[a a_a U_r + a'_a V_r], \quad \frac{1}{p}(p^3 b_{ap+r} - b_a a_r) \equiv 2[a b_a U_r + b'_a V_r] \pmod{p},$$

a form of rank at most two. One channel degenerates identically:

Theorem 2.3. $V_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 (H_{n+k} + H_{n-k} - 2H_k) = 0$ for all $n \geq 0$.

Proof. $V_n = \frac{1}{2} \sum_k \partial_k A_\Gamma(n, k)$ with $A_\Gamma(n, z) = [\Gamma(n+z+1)/(\Gamma(z+1)^2 \Gamma(n-z+1))]^2$, and reflection gives $A_\Gamma(n, z) = (\sin^2 \pi z / \pi^2) g(z)$ with $g(z) = \Gamma(n+z+1)^2 \Gamma(z-n)^2 / \Gamma(z+1)^4$. No analysis is required, because g is *rational*:

$$g = \varphi^2, \quad \varphi(z) = \frac{\prod_{i=1}^n (z+i)}{\prod_{j=0}^n (z-j)}.$$

At an integer $z = k$ the double poles of A_Γ cancel and $\partial_z A_\Gamma|_{z=k} = \text{Res}_{z=k} g$. Now φ has simple poles with residues $\beta_k = (-1)^k \binom{n}{k} \binom{n+k}{k}$ — this is Lagrange interpolation of $\prod_{i=1}^n (X+i)$ at the nodes $0, \dots, n$ — so

$$\text{Res}_{z=m} \varphi^2 = 2\beta_m \sum_{k \neq m} \frac{\beta_k}{m-k},$$

which is antisymmetric in (m, k) and therefore sums to zero over m . $\square$

Remark 2.4. No telescoping proof of Theorem 2.3 of the shape used for the closed form exists: any $\Psi = A \cdot (\alpha H_{n+k} + \beta H_{n-k} + \gamma H_k + \delta)$ forces $r(k)\alpha(k+1) - \alpha(k) = 1$, which is Gosper's equation for the Apéry summand.

Corollary 2.5. For $a, r < p$,

$$\frac{1}{p}(a_{ap+r} - a_a a_r) \equiv 2a a_a U_r, \quad \frac{1}{p}(p^3 b_{ap+r} - b_a a_r) \equiv 2a b_a U_r \pmod{p}.$$

Both rows carry the same functional U_r and differ only by their own value at level a . Equivalently

$$(a_n, p^3 b_n) \equiv (a_a, b_a) \cdot (a_r + 2p a U_r) \pmod{p^2}, \quad n = ap + r.$$

Remark 2.6. U_r is not a_r , nor b_r , nor divisible by a_r : at $p = 5$, $U_r \not\equiv 0$ exactly where $a_r \equiv 0$. It is the exact analogue, for this pair, of the functional appearing in the corresponding law for the Brown–Zudilin $\zeta(5)$ family, and has the same origin, $\partial_n \log$ of the summand.

3. THE SCALAR, AND THE MATRIX

Definition 3.1. With $\Pi(t) = \prod_{i \leq t} (1 + \varepsilon/i)$ and $\bar{\Pi}(t) = \prod_{i \leq t} (1 - \varepsilon/i)$,

$$A_\Gamma(r+\varepsilon, s) = A(r, s) \frac{\Pi(r+s)^2}{\Pi(r-s)^2} \quad (s \leq r), \quad A_\Gamma(r+\varepsilon, r+m) = \varepsilon^2 \left(\frac{(2r+m)!(m-1)!}{((r+m)!)^2} \right)^2 \Pi(2r+m)^2 \bar{\Pi}(m-1)^2,$$

and $\text{Adig}(p, r; \varepsilon) = \sum_{s=0}^{p-1} A_\Gamma(r+\varepsilon, s)$, the sum being over the *full* digit range. Write $X_p(r) = [\varepsilon^2] \text{Adig}(p, r)$.

Then $[\varepsilon^0] = a_r$ and $[\varepsilon^1] = 2U_r$. The terms with digit $s > r$ — the borrow region — are present and are exactly of order ε^2 .

Observation 3.2 ($p = 5, \dots, 23$). With $u(a, r) = a_r + 2p a U_r + p^2 a^2 X_p(r)$,

$$(a_n, p^3 b_n) \equiv (a_a, b_a) \cdot u(a, r) \pmod{p^3},$$

with floor exactly 3. Truncating Adig at orders $m = 0, 1, 2$ gives floors 1, 2, 3, and floor 3 for every $m \geq 3$: the scalar picture saturates at depth three. The same test with the restricted deformation $\sum_{s \leq r}$ saturates at depth two, so the borrow region is exactly the second-order correction.

Observation 3.3 (13 primes, $5 \leq p \leq 47$). The defect of Corollary 2.5 has rank 1 at first order and rank 1 at second order, for both rows, and the two rows have the same r -side row space at every prime. The second-order correction is scalar with a -side factor exactly a^2 .

Proposition 3.4 (the cross term). *Modulo p a full block satisfies $\sum_{t=1}^{p-1} t^{-3} \equiv 0$, so $G_{ap+r} \equiv H_r^{(3)}$ and $G_{cp+s} \equiv H_s^{(3)}$. Hence the piece of $p^3 b_n$ neglected above is $p^3 \sum_k A(n, k)(2G_n - G_k) \equiv p^3 a_a b_r \pmod{p^4}$.*

That is, the low digit acts on the pair by

Conjecture 3.5. For every prime $p \geq 5$ and $n = ap + r$ with $a, r < p$,

$$(a_n, p^3 b_n) \equiv (a_a, b_a) \cdot \begin{pmatrix} u(a, r) & p^3 b_r \\ 0 & u(a, r) \end{pmatrix} \pmod{p^4}, \quad u(a, r) = a_r + 2p a U_r + p^2 a^2 X_p(r).$$

The mod p^2 truncation is Corollary 2.5, proved. The mod p^3 truncation is Observation 3.2. The off-diagonal entry is derived in Proposition 3.4, and is verified at 13 primes in the sharp form: at order p^3 the b -row defect has rank 3, subtracting $p^3 a_a b_r$ drops it to rank 2, and its r -side row space then coincides with the a -row's. The residual scalar defect at p^3 has rank 2 and is not yet named; that is why the full statement is conjectural.

Remark 3.6. The cross entry is the low digit's own companion value, scaled by exactly $p^3 = p^w$ where $w = 3$ is the weight of $\zeta(3)$. So the weight is precisely the order at which the off-diagonal switches on. For the rank-three Brown–Zudilin $\zeta(5)$ family the corresponding statement is that a cross term cannot appear before order p^3 ; here it is derived rather than bounded, and the entry is named.

4. THE REMAINING GAP

Writing $A(ap + r, cp + s) \equiv A(a, c)A(r, s) \exp(2p\Lambda_1 - p^2\Lambda_2)$ with $\Lambda_1 = a u + c v$ and $\Lambda_2 = (a + c)^2 H_{r+s}^{(2)} - 2c^2 H_s^{(2)} - (a - c)^2 H_{r-s}^{(2)}$, the bracket $2\Lambda_1^2 - \Lambda_2$ has three channels:

$$a^2 : 2u^2 - (H_{r+s}^{(2)} - H_{r-s}^{(2)}), \quad ac : 4uv - 2(H_{r+s}^{(2)} + H_{r-s}^{(2)}), \quad c^2 : 2v^2 - (H_{r+s}^{(2)} - 2H_s^{(2)} + H_{r-s}^{(2)}).$$

The measured a -side factor a^2 of Observation 3.3 says the last two die modulo p . They are not identities: $\sum_s A(r, s) \cdot (ac\text{-channel}) = 0, -26, -\frac{905}{2}, -\frac{167965}{18}, \dots$ and the c^2 sums are $0, 11, \frac{607}{4}, \dots$, nonzero over $\mathbb{Q}$ for $1 \leq r \leq 30$. The vanishing is therefore a genuine modulo- p cancellation between the region $s \leq r$ and the borrow region $s > r$ — the same borrow terms that supply the second-order scalar. Proving it is the one gap between Observation 3.3 and a theorem, and the sharpest open question here.

The contrast with the first order is worth recording: there the borrow region contributes nothing at all, because A is a square (Lemma 2.1). At second order it contributes, and its contribution is exactly what kills the two c -channels.